
Cascaded Systems

Silvia Zhang
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Review of Noise and PSDs

PSD:

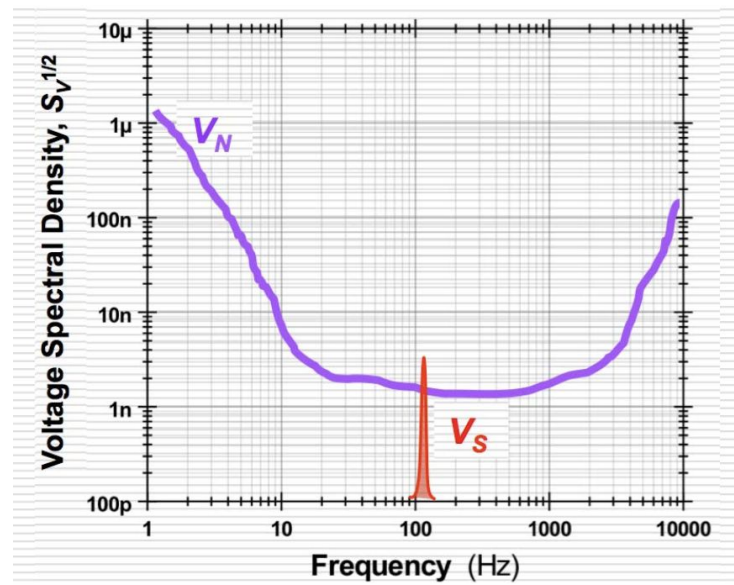
$$S_x(\omega) = |H(\omega)|^2$$

Autocorrelation function:

$$R_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau)dt = \langle x(t)x(t+\tau) \rangle$$

Wiener Khinchin Theorem:

$$S(w) = \int_{-\infty}^{\infty} R_{xx}(\tau)e^{-jw\tau}d\tau$$



Signal + Uncorrelated Noise

If we have a signal with uncorrelated/random noise, we can write them separately:

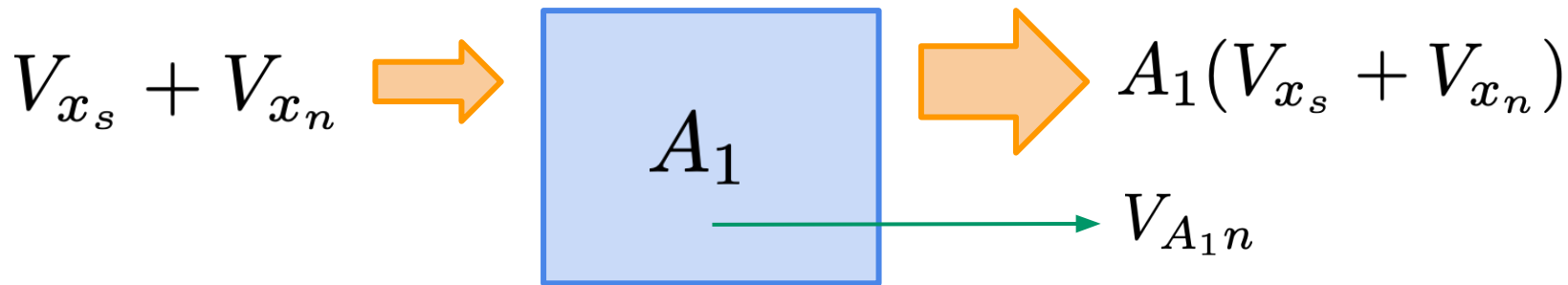
$$V_x = V_{x_s} + V_{x_n}$$

V_x = Voltage Input

V_{x_s} = Voltage Signal

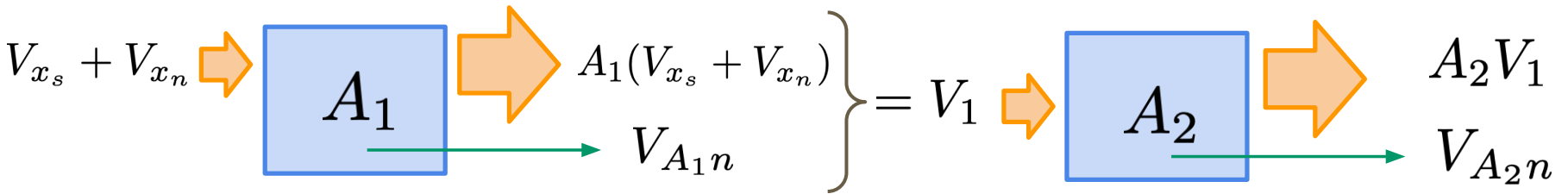
V_{x_n} = Voltage Noise

A Single Amplification Stage



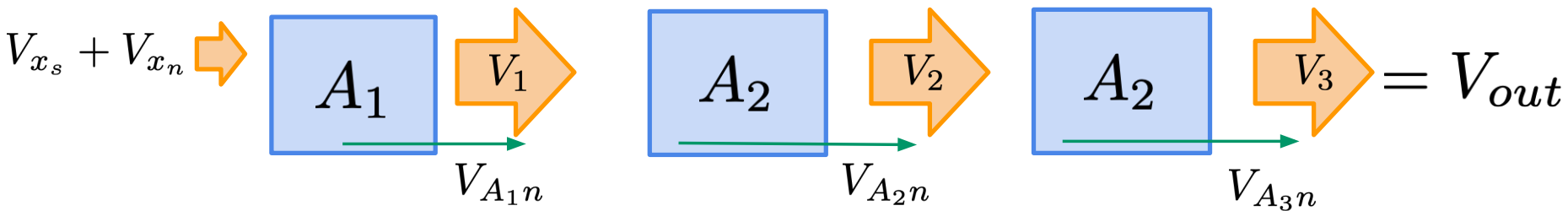
$$V_1 = A_1(V_{x_s} + V_{x_n}) + V_{A_1 n}$$

2 Amplification Stages



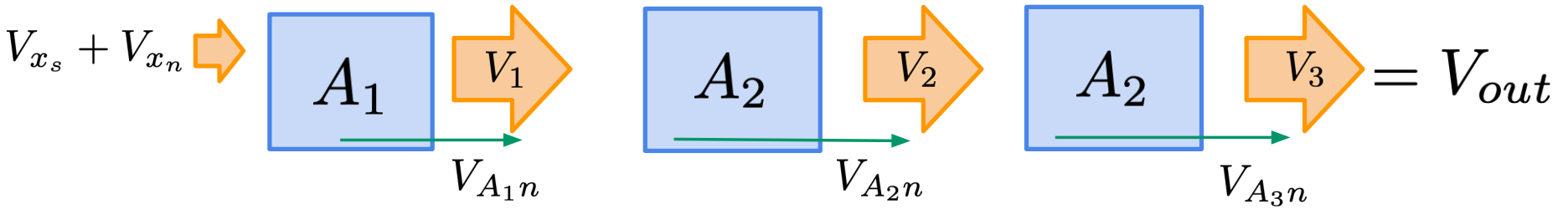
$$V_2 = A_2(A_1(V_{x_s} + V_{x_n}) + V_{A_1n}) + V_{A_2n}$$

3 Amplification Stages



$$V_{out} = V_3 = A_3 A_2 A_1 (V_{x_s} + V_{x_n}) + A_3 A_2 V_{A_1n} + A_3 V_{A_2n} + V_{A_3n}$$

PSD of 3-Stage Cascaded Amplification



$$V_{out} = V_3 = A_3 A_2 A_1 (V_{x_s} + V_{x_n}) + A_3 A_2 V_{A_1n} + A_3 V_{A_2n} + V_{A_3n} \quad [V]$$



$$S_{out} = A_3^2 A_2^2 A_1^2 (S_{x_s} + S_{x_n}) + A_3^2 A_2^2 S_{A_1n} + A_3^2 S_{A_2n} + S_{A_3n} \quad \left[\frac{V^2}{Hz} \right]$$

What if we had Correlated Noise?

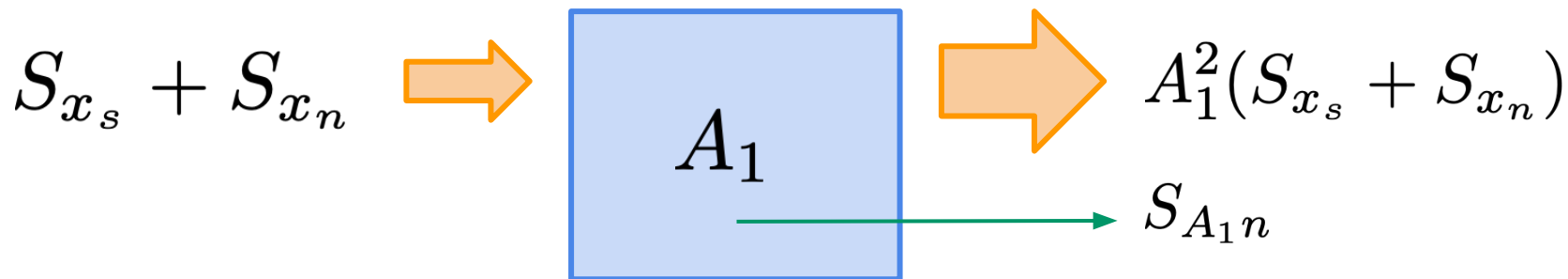
With uncorrelated noise, we only have diagonal terms of the correlation matrix ('auto correlation'):

$$R_{VV}(\tau) = A_3^2 A_2^2 A_1^2 R_{x,x}(\tau) + A_3^2 A_2^2 R_{v_{n1},v_{n1}}(\tau) + A_3^2 R_{v_{n2},v_{n2}}(\tau) + R_{v_{n3},v_{n3}}(\tau)$$

If we had correlated noise, we would have all correlation terms('auto' + 'cross'):

$$R_{VV}(\tau) = \sum_i A_i^2 R_{n_i,n_i}(\tau) + 2 \sum_{i < j} A_i A_j R_{n_i,n_j}(\tau)$$

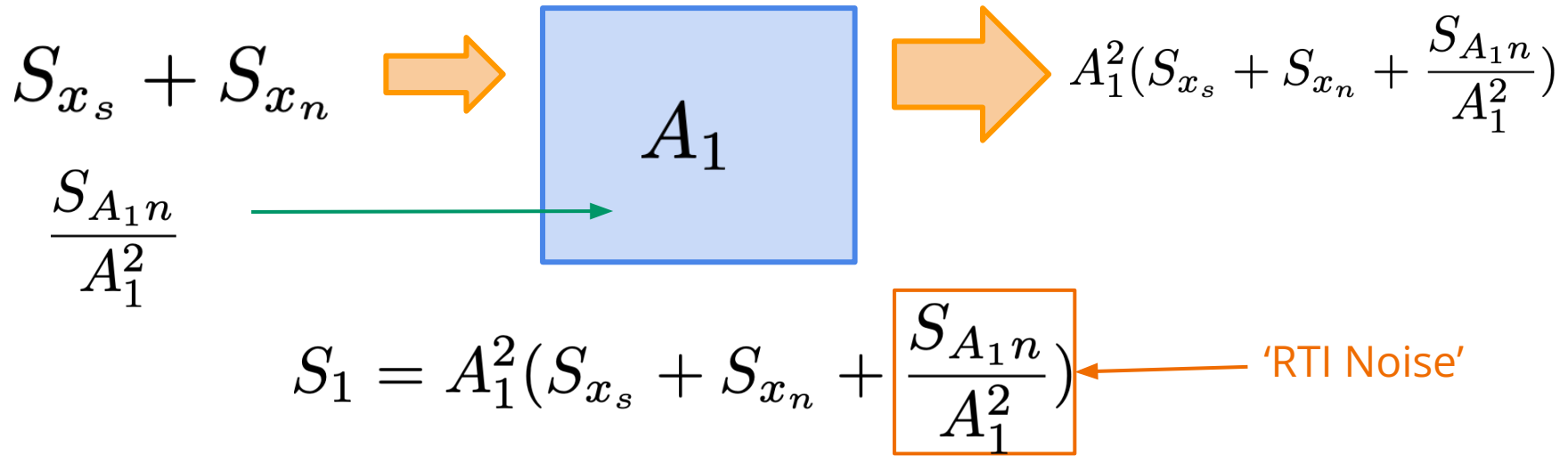
RTO vs. RTI Stage Noise



$$S_1 = A_1^2(S_{x_s} + S_{x_n}) + \boxed{S_{A_1 n}} \leftarrow \text{'RTO Noise'}$$

We can think of the additional noise from an amplifier as if it **adds noise to the output**. This is called “**referenced to output**”.

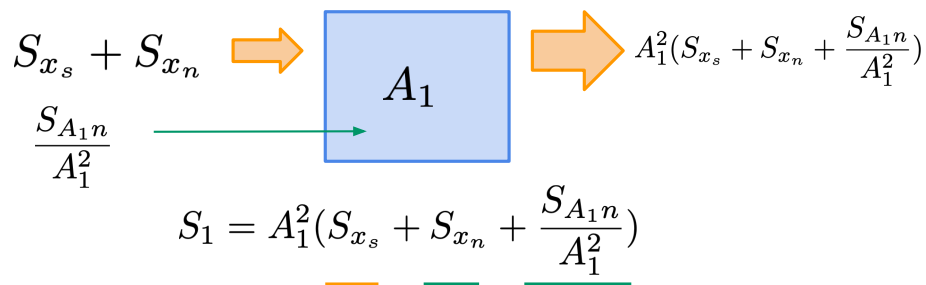
RTO vs. RTI Stage Noise



We can think of the additional noise from an amplifier as if the amplifier itself is **noiseless**, and the **amplifier noise is entering the input**. This is called “**referenced to input**”.

We calculated S_{out} of the system RTO. **What would S_{out} RTI be?**

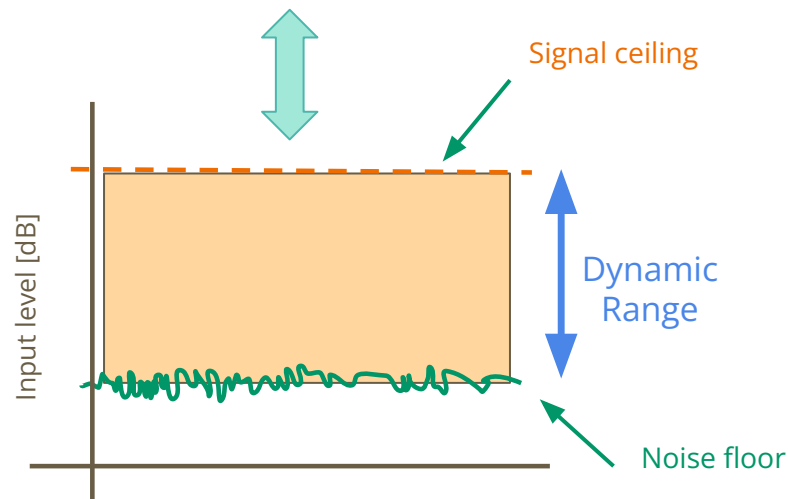
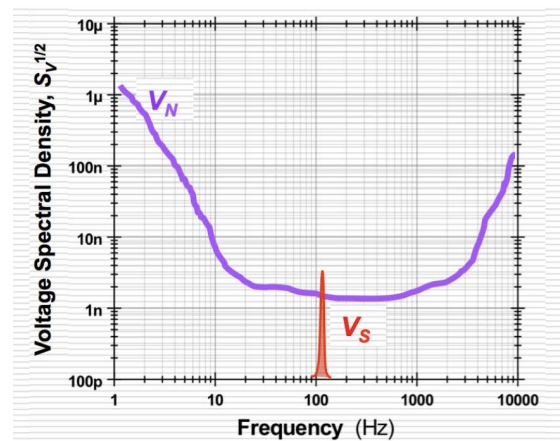
Dynamic Range of a Single Amplifier



We can now think of the input PSD powers in terms of dynamic range.

Maximum signal strength defines the signal ceiling.

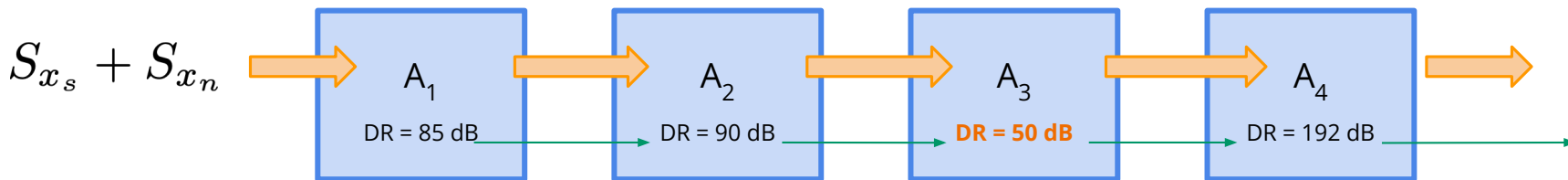
Total noise defines the noise floor.



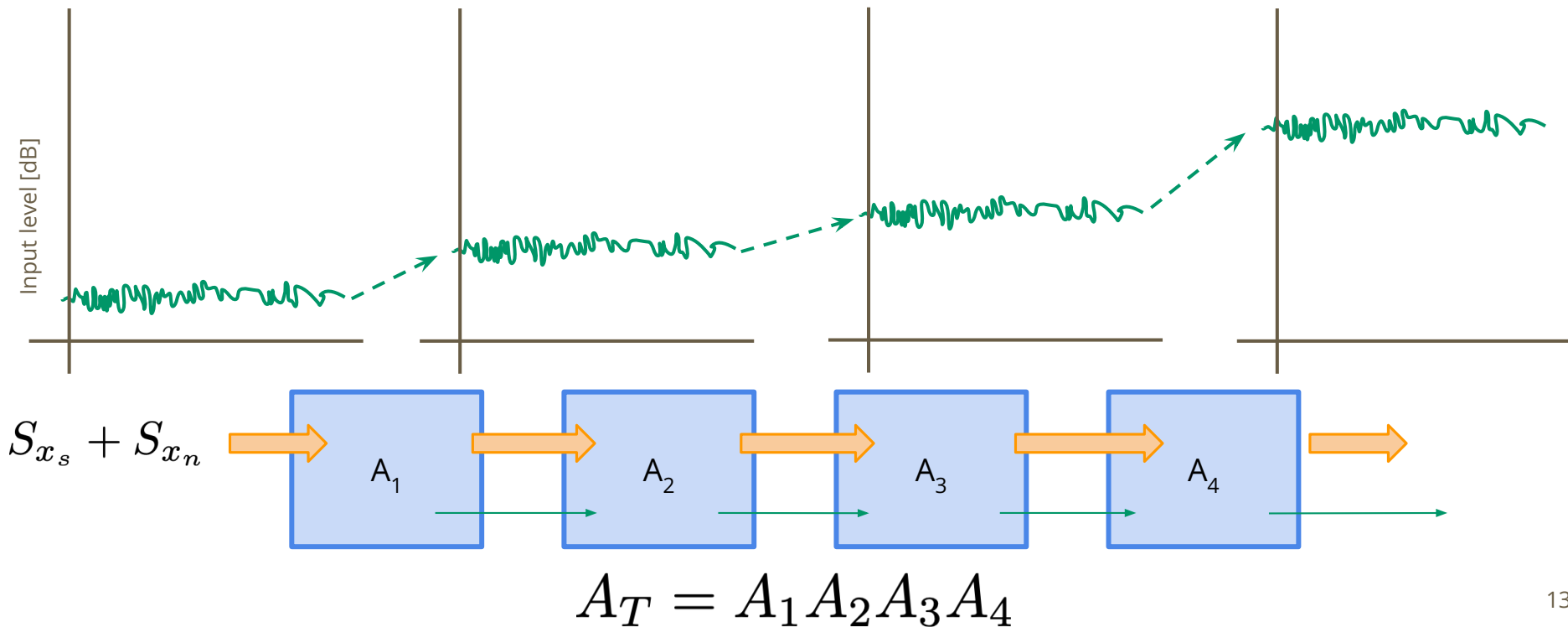
Dynamic Range of a Cascaded System

How will dynamic range change over multiple amplifier stages?

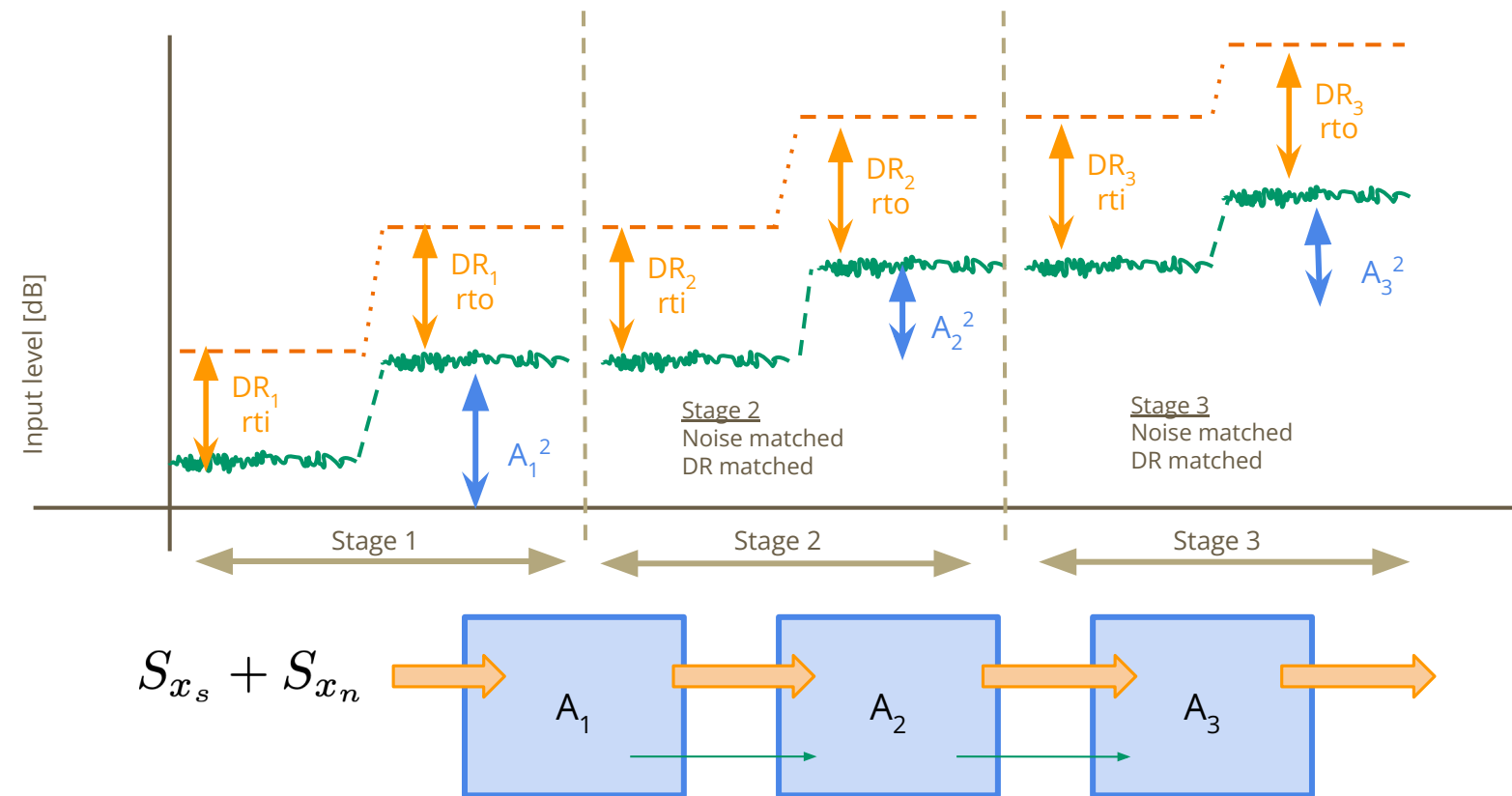
A: The system dynamic range will be set up the least optimal stage.



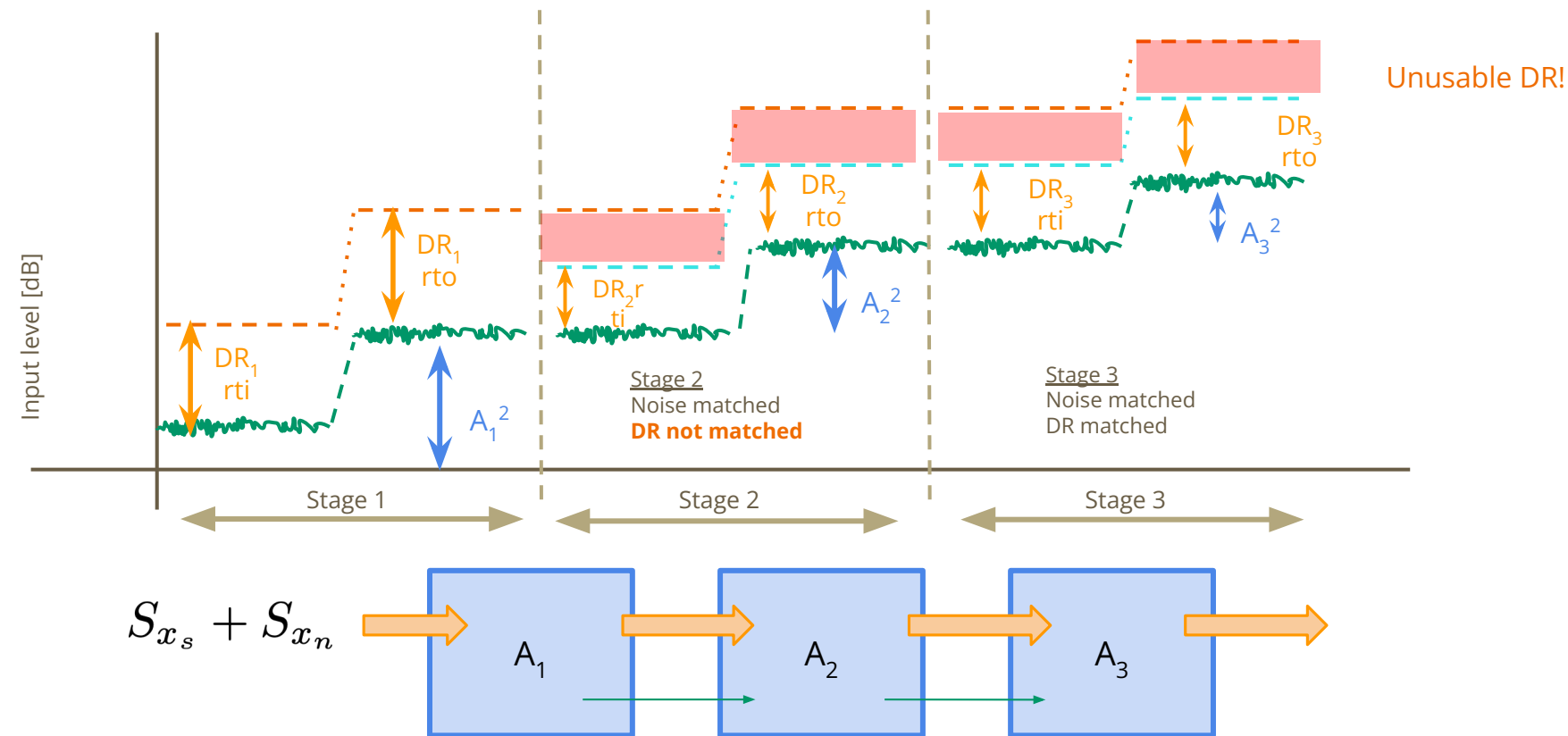
How does Noise Scale over Multiple Amplifiers?



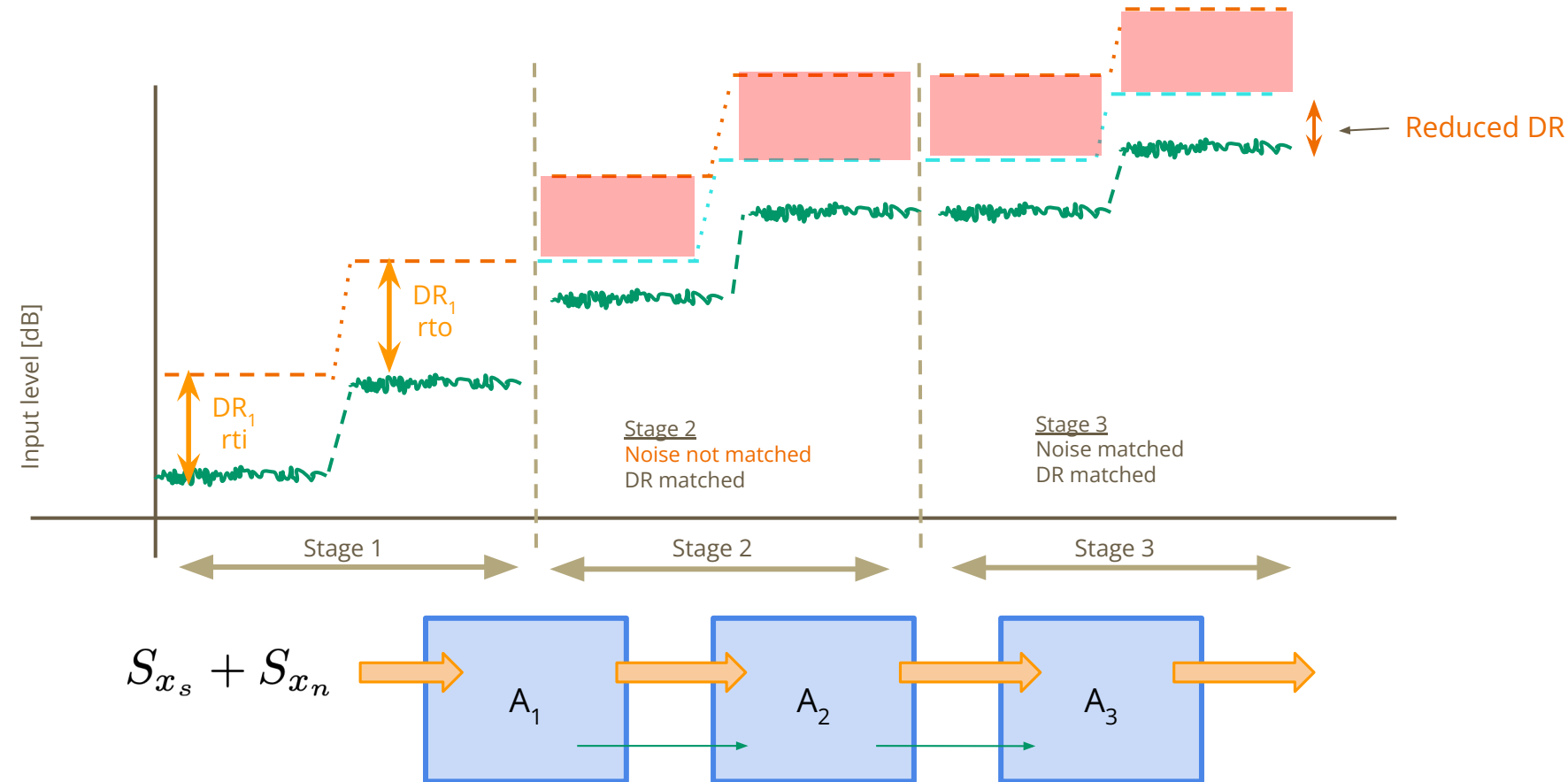
Dynamic Range of an Ideal Cascaded System



Dynamic Range of an Nonideal Cascaded System



Dynamic Range of a Nonideal Cascaded System



Dynamic Range of a Nonideal Cascaded System

